# STALK OF A HIGHER DIRECT IMAGE ALONG A SPECTRAL MAP AS COHOMOLOGY ON THE SET OF SPECIALISATIONS

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*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file cohomology.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* The proof is complete as written and ends with the identification of the colimit with the stalk; no step is referred to the reader.

*Editorial note.* The counted spans carry 2 navigation stubs and no bibliography citation.

*Editorial note.* The preamble is the chapter's retained minimal preamble (sources/stacks-cohomology/retained-preamble.tex lines 1-81), a verbatim excerpt of the native preamble.

*Editorial note.* The label lemma-proper-base-change-spectral is defined only in cohomology.tex, so no cross-chapter restatement had to be weighed for this card.

*Allowed standard prerequisites (editorial).*

- Colimit formula for cohomology on a spectral space presented as an inverse limit. Same chapter, Lemma lemma-colimit, invoked at proof lines 3911-3912. Supplies the identification of the cohomology on the inverse image of E with the colimit over the neighbourhoods.
- Subspaces of spectral spaces appearing as inverse images along spectral maps are spectral. Topology, Lemma topology-lemma-spectral-sub, invoked at proof lines 3905-3906. Supplies that the inverse images of the quasi-compact open neighbourhoods are spectral spaces.
- Stalks of higher direct images as colimits over quasi-compact open neighbourhoods. Standard, used at proof lines 3912-3913. Supplies the right hand side of the identity.

*Source reference.* Original lemma 0A37: sources/stacks-cohomology/pinned-original-source.tex, source lines 3360–3360. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

*Source reference.* Original lemma 0902: sources/stacks-chow/tags.txt, source lines 11395–11395. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

**Lemma 0.1.** *Let $f : X \to Y$ be a spectral map of spectral spaces. Let $y \in Y$. Let $E \subset Y$ be the set of points specializing to $y$. Let $\mathcal{F}$ be an abelian sheaf on $X$. Then $(R^p f_* \mathcal{F})_y = H^p(f^{-1}(E), \mathcal{F}|_{f^{-1}(E)})$.*

*Date*: stacks-project:cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

*Proof.* Observe that $E = \bigcap V$ where $V$ runs over the quasi-compact open neighbourhoods of $y$ in $Y$. Hence $f^{-1}(E) = \bigcap f^{-1}(V)$. This implies that $f^{-1}(E) = \lim f^{-1}(V)$ as topological spaces. Since $f$ is spectral, each $f^{-1}(V)$ is a spectral space too (Topology, Lemma 0902). We conclude that $f^{-1}(E)$ is a spectral space and that

$$H^p(f^{-1}(E), \mathcal{F}|_{f^{-1}(E)}) = \operatorname{colim} H^p(f^{-1}(V), \mathcal{F})$$

by Lemma 0A37. On the other hand, the stalk of $R^p f_* \mathcal{F}$ at $y$ is given by the colimit on the right. $\square$